Passthrough-Style Simulation Equations

Quest Shooting Demo

1 Overview

This document describes the project's screen-space shader and its head-motion controller. It modifies a rendered virtual scene; it does not access the Quest camera feed. The effect is an artistic approximation, not a calibrated model of Quest hardware.

All vector operations below are component-wise unless stated otherwise. Screen coordinates $\mathbf{u} = (u, v)$ range from 0 to 1, and t is elapsed time in seconds.

2 View center and edge weight

Let $\mathbf{c}$ be the projected forward view center for the current rendering view, obtained from the projection matrix in the vertex shader. For each pixel,

$$\mathbf{p} = \mathbf{u} - \mathbf{c}, \qquad r = \|\mathbf{p}\|_2, \qquad e = \text{smoothstep}(0.2, 0.75, r). \tag{1}$$

The edge weight e increases from zero toward one away from the view center.

3 Geometric warp

With warp strength $k_w = 0.035$ and edge warp boost $k_e = 1.4$ by default,

$$w = 1 + k_w r^2 (1 + k_e e), \tag{2}$$

$$\mathbf{u}_0 = \mathbf{c} + \mathbf{p} w, \tag{3}$$

$$\Delta\mathbf{u}_{\text{edge}} = 0.6\, k_w e \mathbf{p}. \tag{4}$$

The edge displacement is added after the swim offsets.

4 Head motion and animated swim

The controller compares camera orientations on consecutive frames:

$$R_\Delta = R_{\text{previous}}^{-1} R_{\text{current}}, \tag{5}$$

$$a = \|\text{Euler}(R_\Delta)\|_2, \tag{6}$$

$$m = \text{clamp}(a\, s, 0, 1), \tag{7}$$

$$h_{\text{new}} = \text{lerp}\left(h_{\text{previous}}, m, 1 - e^{-\Delta t/\tau}\right). \tag{8}$$

Here $s = 4.0$ is the default motion sensitivity, $\tau = 0.12$ seconds is the default smoothing time, and Δt is the frame duration. The angle change is measured per frame, rather than divided by Δt; consequently the response also depends on frame rate.

With swim strength $k_s = 0.018$ by default, $h = h_{\text{new}}$, and $T = 1.7t$, the shader calculates:

$$A = k_s(0.4 + 1.6h), \tag{9}$$

$$u_1 = u_{0x} + A(0.3 + e)\sin(12u_{0y} + T), \tag{10}$$

$$v_1 = u_{0y} + A(0.3 + 0.8e)\cos(11u_1 + 0.9T), \tag{11}$$

$$\mathbf{u}' = (u_1, v_1) + \Delta\mathbf{u}_{\text{edge}}. \tag{12}$$

The vertical offset uses the already modified horizontal coordinate u_1. The $0.4k_s$ term causes swim even when the head is still. Motion here represents rotation only, not translation or scene depth.

5 Softness

Let $\mathbf{D} = (W, H)$ be the viewport size in pixels, $b = 0.7$ the default softness in pixels, and $\mathbf{d} = (b/W, b/H)$. Let $I(\mathbf{x})$ be the screen texture sampled at $\mathbf{x}$ after clamping each coordinate to $[0.001, 0.999]$. The blurred color is

$$\begin{aligned} B = {} & 0.4I(\mathbf{u}') \\ & + 0.15I(\mathbf{u}' + (d_x, 0)) + 0.15I(\mathbf{u}' - (d_x, 0)) \\ & + 0.15I(\mathbf{u}' + (0, d_y)) + 0.15I(\mathbf{u}' - (0, d_y)). \end{aligned} \tag{13}$$

The five weights sum to one.

6 Grain

The luminance of the blurred color $B = (B_R, B_G, B_B)$ is

$$L = 0.2126B_R + 0.7152B_G + 0.0722B_B. \tag{14}$$

The shader constructs a pseudo-random noise value using

$$\mathbf{q} = \lfloor \mathbf{u}\mathbf{D} \rfloor + \lfloor 24t \rfloor, \tag{15}$$

$$N = \operatorname{fract}\big(\sin\big(\mathbf{q} \cdot (12.9898, 78.233)\big)\, 43758.5453\big) - 0.5. \tag{16}$$

The scalar $\lfloor 24t \rfloor$ is added to both components of $\mathbf{q}$. With grain strength $k_g = 0.015$ by default, the resulting color is

$$G = B + N\, k_g\big(1.25 - \operatorname{clamp}(L, 0, 1)\big). \tag{17}$$

The noise is stronger in darker rendered regions; it does not measure physical lighting conditions.

7 Edge darkening and output

With edge-darkening strength $k_v = 0.2$ by default,

$$V = \operatorname{smoothstep}(0.45, 0.71, r), \tag{18}$$

$$C_{\text{out}} = \operatorname{clamp}\big(G(1 - k_v V), 0, 1\big). \tag{19}$$

The final clamp is applied separately to each RGB component. The vignette suggests a limited view but does not change the headset's physical field of view.

8 Pipeline

$$\boxed{\text{Output} = \text{edge darkening}\big(\text{softened, warped scene} + \text{grain}\big)}$$

The screen texture contains opaque scene rendering; transparent objects and UI may not receive the effect.